

418hw3

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Problem 1

```
n1 = 20
alpha = 0.05
means1 = matrix(c(4.64,45.4,9.965),3,1)
dispersion1 = matrix(c(2.879,10.01,-1.81,10.01,199.788,-5.64,-1.81,-5.64,3.628),
                     3,3,byrow=TRUE)
mu_nought1 = matrix(c(4,65,10),3,1)
test_stat1 = n1*t(means1-mu_nought1) %*% solve(dispersion1) %*%
             (means1-mu_nought1)
crit_val1 = qchisq(alpha,length(means1),lower.tail=FALSE)
test_stat1 > crit_val1
```

```
##      [,1]
## [1,] TRUE
```

```
# The above logical gives "true", meaning we reject the null hypothesis and
# conclude that at least one of the variable means is significantly different
# to the hypothesized means. This would appear the sodium content hypothesized
# mean, because it is furthest from its respective sample mean, and said
# variable has the largest variance by a considerable margin.
```

Problem 2

```
p2_data = read.csv("C:/Users/jacma/Downloads/T3_5_probe.csv")
# a
n2 = 11
mu_nought2 = matrix(c(30,25,40,25,30))
raw_data = p2_data[,2:6] # extract needed info
means2 = colMeans(raw_data) # sample means
dispersion2 = cov(raw_data) # in-built covariance function for dispersion matrix
test_stat2 = n2*t(means2-mu_nought2) %*% solve(dispersion2) %*%
             (means2-mu_nought2)
crit_val2 = qchisq(alpha,length(means2),lower.tail=FALSE)
test_stat2 > crit_val2
```

```
##      [,1]
## [1,] TRUE

# Logical gives "True", so we reject the null hypothesis and conclude that at
# least one mean is significantly different to its respective hypothesized value
# b
for (i in 1:5) { # loop through each of the 5 means (5 variables)
  zsq = (n2*(means2[i] - mu_nought2[i])^2)/dispersion2[i,i]
  critical = qchisq(alpha,1,lower.tail=FALSE)
  cat(i,'Chi:',abs(zsq) > critical,'\n') # print decision
  z = sqrt(n2)*(means2[i]-mu_nought2[i])/sqrt(dispersion2[i,i])
  cat(i,'Norm:',abs(z) > qnorm(1-alpha/2),'\n')
}

## 1 Chi: TRUE
## 1 Norm: TRUE
## 2 Chi: FALSE
## 2 Norm: FALSE
## 3 Chi: TRUE
## 3 Norm: TRUE
## 4 Chi: FALSE
## 4 Norm: FALSE
## 5 Chi: FALSE
## 5 Norm: FALSE
```

```
# We reject null hypothesis only for the first and third means, indicating the
# first and third probe word means are significantly different to their
# hypothesized value. From this, it can be said that these variables contributed
# most to the rejection of the overall hypothesis, but the increase type 1 error
# as a result of this procedure should also be considered.
```

Problem 3

```
p3_data = read.csv("C:/Users/jacma/Downloads/T5_5_fleaBeetles.csv")
group1 = p3_data[1:19,3:6] # data calculation
group2 = p3_data[20:39,3:6]
n1 = 19 # group 1 info
g1_means = colMeans(group1)
g1_disp = cov(group1)
n2 = 20 # group 2 info
g2_means = colMeans(group2)
g2_disp = cov(group2)
# a
# pool variances
spooled = ((n1-1)*g1_disp + (n2-1)*g2_disp)/(n1+n2-2)
tsq = (n1*n2/(n1+n2))*t(g1_means-g2_means)%*%solve(spooled)%*(g1_means-g2_means)
# critical value: T^2_0.05,4,37 or convert to F...
factor = (n1+n2-5)/(4*(n1+n2-2))
adj_t = factor*tsq # df1 = p, df2 = n1+n2-p-1
cat('Mean vector comparison p-value:', pf(adj_t,4,34,lower.tail=FALSE),'\n')
```

```
## Mean vector comparison p-value: 7.521799e-11
```

```
# The p-value is significantly smaller than alpha = 0.05, so we reject the null  
# hypothesis and conclude that the mean of at least one variable is different  
# between the two groups
```

```
library(Hotelling) # confirm with Hotelling test
```

```
## Loading required package: corpcor
```

```
result = hotelling.test(group1,group2)  
result
```

```
## Test stat: 133.49  
## Numerator df: 4  
## Denominator df: 34  
## P-value: 7.522e-11
```

```
# b  
crit_t = abs(qt(alpha/2,n1+n2-2)) # critical value, constant  
for (i in 1:4) { # loop through the 4 variables  
  pooled_var = ((n1-1)*g1_disp[i,i]+(n2-1)*g2_disp[i,i])/(n1+n2-2)  
  test_t = (g1_means[i]-g2_means[i])/sqrt(pooled_var*(1/n1+1/n2)) # test stat  
  cat(i,abs(test_t) > crit_t,'\n') # logical operator, print out  
}
```

```
## 1 TRUE  
## 2 TRUE  
## 3 TRUE  
## 4 TRUE
```

```
# TRUE representing a rejection of H_0, it can be seen that all of the variable  
# means in one group are significantly different to the corresponding variable  
# means in the other group. This explains the extremely small p-value obtained  
# in the multivariate test.
```

```
# c  
a1 = solve(spoiled) %*% (g1_means-g2_means)  
cat('Discriminant coefficient vector:', a1)
```

```
## Discriminant coefficient vector: 0.345249 -0.1303878 -0.1064338 -0.1433533
```

```
# The first variable, y1 = distance of transverse groove from posterior border,  
# appears to be the most significant based on the discriminant vector because  
# its coefficient is the largest in absolute value. The other three variables  
# seem to be comparable in terms of effect, as their coefficients are close  
# together in magnitude.
```

Problem 4

```

p4_data = read.csv('C:/Users/jacma/Downloads/T5_9_essay.csv') # load data
# a
n = 15
d1 = p4_data[,2]-p4_data[,4] # calculate mean differences
d2 = p4_data[,3]-p4_data[,5]
differences = data.frame(d1,d2)
d_means = colMeans(differences)
s_d = cov(differences)
test_stat3 = n*t(d_means)%*%solve(s_d)%*%d_means
factor = (n1+n2-5)/(4*(n1+n2-2))
adj_test = factor*tsq # df1 = p, df2 = n-p
cat('Mean vector comparison p-value:', pf(adj_test,2,13,lower.tail=FALSE),'\n')

```

```
## Mean vector comparison p-value: 1.19671e-05
```

```

# The p-value is lower than alpha = 0.05, so we reject the null hypothesis and
# conclude that at least one of the difference means in the vector is
# significantly different to 0.

```

```

# b
a2 = solve(s_d)%*%d_means
cat('Discriminant coefficient vector:', a2,'\n')

```

```
## Discriminant coefficient vector: -0.03601558 0.04770629
```

```

# The two coefficients are relatively close in absolute value, but the d2
# coefficient is slightly larger, meaning the mean difference between verbs
# in an informal essay versus a formal essay is more significant in the rejection
# of the null hypothesis. That is, the number of verbs is a more significant
# change between formal and informal essays than the total number of words.
# This result could be because the difference in verbs has a lower variance,
# though this can be expected because there are inherently less verbs than there
# are total words, leading to smaller observation values, in turn leading to a
# smaller variance.

```

```

# c
uni_crit = abs(qt(alpha/2,n-1))
for (i in 1:2) { # test each mean difference
  uni_test = sqrt(n)*d_means[i]/(sqrt(s_d[i,i]))
  cat(i,abs(uni_test) > uni_crit,'\n')
}

```

```
## 1 TRUE
```

```
## 2 TRUE
```

```

# The null hypothesis that  $\mu_{di} = 0$  was rejected for both  $i=1$  and  $i=2$ , meaning
# both mean differences are significant different to 0. This suggests that there
# is a difference between formal and informal essays with respect to verb count
# AND total word count, but the increase in Type 1 error from this procedure
# should be considered.

```